

Question number	Answer	Additional guidance	Mark
6(a)	A is the correct answer • B is not the correct answer as Pr does not break down in the dark • C is not the correct answer as Pr does not absorb far red light • D is not the correct answer as Pr does not absorb far red light nor break down in the dark		Computer (1)

Question number	Answer	Additional guidance	Mark
6(b)	An explanation that includes three of the following points: - auxin accumulates on shaded side (of the tip/stem/shoot/plant) (1) {stimulating/causing} cell elongation (1) - detail of mechanism of action of auxin (1) - causing the plant to grow towards {light / sun} (1)	accept accumulates on the left of diagram accept auxins move away from the exposed light e.g. auxins stimulate the transport of Hydrogen ions into the cell wall or decrease pH activating {cell wall proteins / enzymes / chemicals / expansins} / {altering the hydrogen bonding within / decreasing strength of} the cell wall allowing elongation when water taken in / ref to activation of transcription factors accept causing positive phototropism do not accept bend towards light	Expert (3)

Question number	Answer	Additional guidance	Mark
6(c)	An explanation that includes four of the following points: • (gibberellin) binds to receptor / carrier molecule in the cell / acts as a transcription factor(1) • (amylase) {transcription factor/gene is activated} (1) • allowing {(amylase gene) transcription / production of (amylase) mRNA / translation mRNA / production of amylase} (1) • amylase (hydrolyses/ converts) starch into {maltose / glucose} (1)	ignore amylose transcription factor binding to promotor region of gene cause the inactivation of (amylase) gene {inhibitor protein/repressor molecule} ignore monomers activates gene that produces amylase = mps 2 and 3	Expert (4)

Question number	Answer	Additional guidance	Mark
6(d)	An answer that includes four of the following points: • both natural and synthetic auxin cause an increase (in cell number) (1) • {natural auxin/ IAA} greater effect on growth than {synthetic auxin / NAA} (1) • IAA at $0.1 \mu\text{mol dm}^{-3}$ was the most effective at increasing {growth / cell number} (1) • $1 \mu\text{mol dm}^{-3}$ was the most effective NAA concentration at increasing {growth / cell number} (1) • IAA is {better / more effective / has more cells} at lower concentrations(1)	can piece together both natural and synthetic auxin are better than the control at all concentrations IAA gives a greater increase in number of cells than NAA Accept most cell number Accept most cell number Accept higher concentrations of auxin(IAA) {had lower increase in growth / had lower increase in number of cells} Accept converse Accept negative correlation	Expert (4)

